# Optimization — Homework 4

Worked practice solutions

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*Scope.* Problems 1–4 and optional Problem 5 from the supplied handout. Vector inequalities are componentwise; $\mathbf{1}$ is an all-ones vector of the required dimension. These are companion explanations for practice.

## Problem 1. A nonnegative invariant vector

Let $P \in \mathbb{R}^{n\times n}$ be row stochastic: $P \geq 0$ and $P\mathbf{1} = \mathbf{1}$. We seek a nonzero $\lambda \geq 0$ with $(P^{\mathsf{T}} - I)\lambda = 0$.

*Proof.* Consider the auxiliary primal–dual pair

$$\begin{array}{lll} \text{maximize} & \mathbf{1}^{\mathsf{T}}\lambda & \\ \text{subject to} & (P^{\mathsf{T}} - I)\lambda = 0 & \\ & \lambda \geq 0 & \end{array} \qquad \begin{array}{lll} \text{minimize} & 0 & \\ \text{subject to} & (P - I)y \geq \mathbf{1}, & \\ & y \in \mathbb{R}^n. & \end{array}$$

The dual follows from the standard pair $\max\{d^{\mathsf{T}}z : Bz = h, z \geq 0\}$ and $\min\{h^{\mathsf{T}}y : B^{\mathsf{T}}y \geq d, y \text{ free}\}$.

The dual is infeasible. Indeed, for any $y$, choose $k$ such that $y_k = \max_j y_j$. Nonnegativity and unit row sums give

$$(Py)_k = \sum_j p_{kj}y_j \leq \sum_j p_{kj}y_k = y_k,$$

so $((P - I)y)_k \leq 0$, contradicting the required lower bound 1.

The primal is feasible at $\lambda = 0$. If it had no nonzero feasible point, its optimum would be the finite value 0; strong LP duality would then give a feasible dual optimum. This contradicts dual infeasibility. Therefore a nonzero feasible $\lambda$ exists.

Since $\lambda \geq 0$ and $\lambda \neq 0$, its sum is positive. Hence

$$\pi = \frac{\lambda}{\mathbf{1}^{\mathsf{T}}\lambda} \quad \Longrightarrow \quad P^{\mathsf{T}}\pi = \pi, \qquad \pi \geq 0, \qquad \mathbf{1}^{\mathsf{T}}\pi = 1.$$

Thus $\pi$ is an invariant probability distribution. □

**Result.** Every finite row-stochastic matrix has a nonnegative left eigenvector for eigenvalue 1, which can be normalized to a probability vector.

**Interpretation.** $P\mathbf{1} = \mathbf{1}$ already gives a right eigenvector; the proof supplies the nonnegative *left* eigenvector needed for a column-vector probability distribution. It does not assert uniqueness or convergence of the Markov chain.

## Problem 2. A piecewise-linear convex objective

Write

$$f(x) = \max_{1 \le i \le m} (a_i^\mathsf{T} x - b_i), \qquad v = \inf_{x \in \mathbb{R}^n} f(x) \in \mathbb{R},$$

where $m \ge 1$, $A \in \mathbb{R}^{m \times n}$ has rows $a_i^\mathsf{T}$, and $b \in \mathbb{R}^m$.

### (a) Every feasible weight vector gives a lower bound

*Proof.* Let $p \ge 0$, $\mathbf{1}^\mathsf{T} p = 1$, and $A^\mathsf{T} p = 0$. A weighted average is no larger than the largest entry, so for every $x$,

$$f(x) \ge \sum_i p_i (a_i^\mathsf{T} x - b_i) = x^\mathsf{T} A^\mathsf{T} p - p^\mathsf{T} b = -p^\mathsf{T} b.$$

Taking the infimum over $x$ proves $-p^\mathsf{T} b \le v$. □

### (b) The best lower bound is exact

*Proof.* Introduce a free scalar $t$ to obtain the equivalent epigraph LP

$$\begin{aligned} \min_{x \in \mathbb{R}^n,\ t \in \mathbb{R}} \quad & t \\ \text{s.t.} \quad & Ax - t\mathbf{1} \le b. \end{aligned}$$

For fixed $x$, the smallest feasible $t$ is exactly $f(x)$. For the inequality $Ax - t\mathbf{1} - b \le 0$, use multipliers $p \ge 0$:

$$\begin{aligned} L(x, t, p) &= t + p^\mathsf{T}(Ax - t\mathbf{1} - b) \\ &= (A^\mathsf{T} p)^\mathsf{T} x + (1 - \mathbf{1}^\mathsf{T} p) t - p^\mathsf{T} b. \end{aligned}$$

Because $x$ and $t$ are unrestricted, the infimum of $L$ over them is finite precisely when $A^\mathsf{T} p = 0$ and $\mathbf{1}^\mathsf{T} p = 1$. It is then $-p^\mathsf{T} b$. The dual is therefore

$$\boxed{\max_{p} \ -p^\mathsf{T} b \quad \text{s.t.} \quad A^\mathsf{T} p = 0, \quad \mathbf{1}^\mathsf{T} p = 1, \quad p \ge 0.}$$

The epigraph LP is feasible, for example at $x = 0$, $t = \max_i(-b_i)$, and its objective is bounded below by the assumed finite value $v$. A feasible LP with a finite optimal value attains it. Strong duality gives an attained dual optimum equal to $v$. □

**Connection to Homework 3.** The dual constructs a certificate by combining constraint rows with nonnegative weights. Here the weights sum to one because the coefficient of the free epigraph variable must vanish. At an optimal pair, complementary slackness gives

$$p_i^* [t^* - (a_i^\mathsf{T} x^* - b_i)] = 0.$$

Only affine pieces that attain the maximum at $x^*$ can receive positive weight. Their slopes balance: $\sum_i p_i^* a_i = 0$.

## Problem 3. Symmetry produces an optimality certificate

The primal and dual are

$$\begin{array}{lll} \text{minimize} & c^{\mathsf{T}}x & \\ \text{subject to} & Ax \geq c & \\ & x \geq 0 & \end{array} \qquad \begin{array}{lll} \text{maximize} & c^{\mathsf{T}}y & \\ \text{subject to} & A^{\mathsf{T}}y \leq c, & \\ & y \geq 0. & \end{array}$$

*Proof.* Suppose $Ax^* = c$ and $x^* \geq 0$. Then $x^*$ is primal feasible. Since $A = A^{\mathsf{T}}$, the choice $y^* = x^*$ also satisfies $A^{\mathsf{T}}y^* = c \leq c$ and $y^* \geq 0$, so it is dual feasible. Their objective values coincide: $c^{\mathsf{T}}x^* = c^{\mathsf{T}}y^*$. Weak duality therefore certifies that both are optimal. □

**The same certificate without dual notation.** For any feasible $x$,

$$c^{\mathsf{T}}x = (Ax^*)^{\mathsf{T}}x = (x^*)^{\mathsf{T}}Ax \geq (x^*)^{\mathsf{T}}c = c^{\mathsf{T}}x^*.$$

The middle equality uses symmetry; the inequality uses $x^* \geq 0$ and $Ax \geq c$. No positive-semidefinite assumption on $A$ is needed.

## Problem 4. Robust right-hand-side uncertainty

The displayed uncertainty set is

$$\mathcal{U} = \{b \in \mathbb{R}^m : 0 \leq b_i \leq 1,\ \sum_i b_i \leq 1\}.$$

*Notation note.* Although the handout calls this a unit box, the displayed set is the simplex $\operatorname{conv}\{0, e_1, \ldots, e_m\}$. We use the displayed set; the claimed answer is also valid for $[0,1]^m$.

*Proof.* Suppose $Ax \geq b$ for every $b \in \mathcal{U}$. For each $i$, the coordinate vector $e_i$ belongs to $\mathcal{U}$. Applying the robust constraint to $b = e_i$ yields $(Ax)_i \geq 1$. Thus $Ax \geq \mathbf{1}$.

Conversely, if $Ax \geq \mathbf{1}$, then every $b \in \mathcal{U}$ satisfies $b \leq \mathbf{1}$, and hence $Ax \geq \mathbf{1} \geq b$. The feasible sets coincide and the objective is unchanged, proving equivalence to $\min c^{\mathsf{T}}x$ subject to $Ax \geq \mathbf{1}$. □

**Result.** The robust right-hand side is the vector of coordinatewise maxima: $\widehat{b}_i = \max_{b \in \mathcal{U}} b_i = 1$.

For $m > 1$, $\mathbf{1} \notin \mathcal{U}$. This causes no problem: different rows can have different worst-case scenarios. There is no sign restriction on $x$ in this problem unless separately imposed.

## Problem 5 (optional). Finite robust counterparts

**Reading the handout.** We use $Ax \le b$ with $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$, consistent with its statement that there are $m$ constraints. The printed $A^\mathsf{T}x$ is dimensionally inconsistent with those dimensions. Equivalently, one can keep $A^\mathsf{T}x$ and take $A \in \mathbb{R}^{n\times m}$. We call the last uncertainty set $\mathcal{U}_3$, correcting its repeated $\mathcal{U}_2$ label.

Let

$$\mathcal{U}_2 = \{a \in \mathbb{R}^n : D_2 a = d_2,\ a \ge 0\}, \qquad \mathcal{U}_3 = \{a \in \mathbb{R}^n : D_3 a \ge d_3,\ a \le 0\},$$

where $D_j \in \mathbb{R}^{r_j \times n}$ and $d_j \in \mathbb{R}^{r_j}$. First assume all uncertainty sets are nonempty.

### 1. Uncertain right-hand side

Since $\mathcal{U}_1 \subseteq \mathbb{R}^m$ is a nonempty polytope, compute the finite constants

$$\underline{b}_i = \min_{b\in\mathcal{U}_1} b_i, \qquad i = 1, \dots, m.$$

Then $Ax \le b$ for every $b \in \mathcal{U}_1$ is equivalent to $Ax \le \underline{b}$. If $\mathcal{U}_1 = \{b : Gb \le h\}$, each $\underline{b}_i$ is the value of the LP $\min\{e_i^\mathsf{T} b : Gb \le h\}$. If vertices $v^1, \dots, v^K$ are given instead, use $\underline{b}_i = \min_k v_i^k$.

### 2. Coefficients in $\mathcal{U}_2$

For fixed $x$, the worst-case left-hand side is

$$\begin{aligned} s_2(x) &= \sup\{x^\mathsf{T} a : D_2 a = d_2,\ a \ge 0\} \\ &= \min\{d_2^\mathsf{T} y_2 : D_2^\mathsf{T} y_2 \ge x,\ y_2 \text{ free}\}, \end{aligned}$$

whenever the supremum is finite, by LP strong duality. Thus the robust inequality is equivalent to the existence of $y_2$ with

$$\boxed{D_2^\mathsf{T} y_2 \ge x, \qquad d_2^\mathsf{T} y_2 \le b_2, \qquad y_2 \in \mathbb{R}^{r_2}.}$$

*Why this is exact.* If such $y_2$ exists, every $a \in \mathcal{U}_2$ satisfies $x^\mathsf{T} a \le y_2^\mathsf{T} D_2 a = y_2^\mathsf{T} d_2 \le b_2$. Conversely, if the robust inequality holds, the inner maximization is nonempty and bounded above by $b_2$. Its dual therefore has an optimizer that satisfies the displayed conditions. □

### 3. Coefficients in $\mathcal{U}_3$

Set $z = -a \geq 0$. Then $D_3 a \geq d_3$ becomes $D_3 z \leq -d_3$, and $x^\mathsf{T} a = (-x)^\mathsf{T} z$. The worst-case LP and its dual are

$$\begin{aligned} s_3(x) &= \sup\{(-x)^\mathsf{T} z : D_3 z \leq -d_3,\ z \geq 0\} \\ &= \min\{-d_3^\mathsf{T} \mu_3 : D_3^\mathsf{T} \mu_3 \geq -x,\ \mu_3 \geq 0\}, \end{aligned}$$

when the supremum is finite. Consequently,

$$\boxed{D_3^\mathsf{T} \mu_3 \geq -x, \qquad -d_3^\mathsf{T} \mu_3 \leq b_3, \qquad \mu_3 \geq 0.}$$

*Direct sign check.* For any $a \in \mathcal{U}_3$ and such a $\mu_3$, multiplying $D_3^\mathsf{T} \mu_3 \geq -x$ by $a \leq 0$ reverses the inequality, giving $a^\mathsf{T} D_3^\mathsf{T} \mu_3 \leq -a^\mathsf{T} x$. Also $\mu_3^\mathsf{T} D_3 a \geq \mu_3^\mathsf{T} d_3$. Therefore $x^\mathsf{T} a \leq -\mu_3^\mathsf{T} D_3 a \leq -\mu_3^\mathsf{T} d_3 \leq b_3$. Necessity follows from strong duality as in the preceding case. □

### The resulting finite LP

For nonempty uncertainty sets, the complete formulation is

$$\boxed{\begin{aligned} \min_{x, y_2, \mu_3} \quad & c^\mathsf{T} x \\ \text{s.t.} \quad & Ax \leq \underline{b}, \\ & D_2^\mathsf{T} y_2 \geq x, \qquad d_2^\mathsf{T} y_2 \leq b_2, \\ & D_3^\mathsf{T} \mu_3 \geq -x, \qquad -d_3^\mathsf{T} \mu_3 \leq b_3, \\ & \mu_3 \geq 0, \qquad x \in \mathbb{R}^n, \quad y_2 \in \mathbb{R}^{r_2}. \end{aligned}}$$

It has finitely many variables and linear constraints. The $m$ constants $\underline{b}_i$ are precomputed by finite LPs.

### Empty and unbounded uncertainty sets

The handout does not explicitly assume nonemptiness. Check each set by an LP feasibility problem first. If a set is empty, its universal constraint is vacuous, so omit the corresponding block above.

$\mathcal{U}_2$ and $\mathcal{U}_3$ need not be bounded. For a given $x$, an unbounded worst-case objective makes its robust constraint impossible for finite $b_j$; the associated dual block then has no feasible certificate. An unbounded *set* need not have an unbounded objective in every direction, so bounded uncertainty is not required for the reformulation.

*Source:* supplied `hw04.pdf`, pp. 1–3. Proofs and explanations are derived here; see the companion study guide for geometric intuition, worked examples, self-tests, and a reference sheet.